

Challenge: Color Sorter

Points: 200 • Estimated Time: 90 Minutes

Challenge Scope

The robot must identify and push three different colored blocks into their corresponding colored drop zones.

Instructions & Engineering Hints

- **Challenge Overview:** Your mission is to build and program a robotic color sorter that can independently identify three differently colored blocks and push each one into its corresponding colored drop zone.
- **Strategic Approach:** Start by dividing the problem into phases: detection, navigation, and sorting. First, focus on getting your robot to reliably detect the color of a block placed in front of it before attempting full movement and sorting.
- **Sensor Placement Tips:** Position your color or light sensor close to the ground for optimal readings, but ensure it is protected from outside ambient light interference by creating a small physical shield or housing around it.
- **Structural Considerations:** Design a reliable pushing mechanism or arm. Consider whether a passive sweep, a motorized plow, or an intake-and-drop system will be most consistent for moving the blocks without knocking them over or missing the target drop zones.
- **Programming Logic:** Utilize conditional statements (if-else logic) in your code. Read the sensor value, evaluate which color range it falls into, and trigger the corresponding movement sequence to route the block to the correct zone.